HUMBER POLYTECHNIC | Faculty of Applied Sciences & Technology

MENG 3050-Power Transmission Components

Assignment 02: Heavy-Duty Scissor Lifts

Introduction:

A customer requested the design of a heavy-duty, automated assembly lift fixture with adjustable height to accommodate operators of varying heights.

The fixture shall:

- Provide adjustable height to accommodate a range of operator heights.
- Operate using standard electrical outlet, 110 VAC, 15Amp.
- Provide a full vertical stroke of 9.75 in. (248 mm).
- Complete the full lifting or lowering stroke within 90 seconds maximum.

Main components:

- Top platform that can be raised straight up and down,
- Scissor mechanism to support and stabilise the platform through raising and collapsing,
- Bottom platform, construct the stationary lower structure, provided with anchoring foots,
- Actuator subassembly, use gearmotor, industry standard electrical outlet,

Scissors lift general information:

- A multi-stage Scissor lifts provide a mobile means of reaching heights that are very high. In the materials handling industry, these include tops of storage racks.
- These machines vary in size. Small lifts, four feet wide, are often used to help navigation within and between tight isles inside warehouses.
- Scissors lifts allow workers to work at areas high above ground level without any concern for balance that they would have to exercise if they were using a ladder.
- One of the basic safety rules forbids operators and users of scissor lifts to move the machine while the platform is in the up position.







Specification:

Draw top, lower platforms, criss-crossing metal supports, and necessary components with the following specifications:

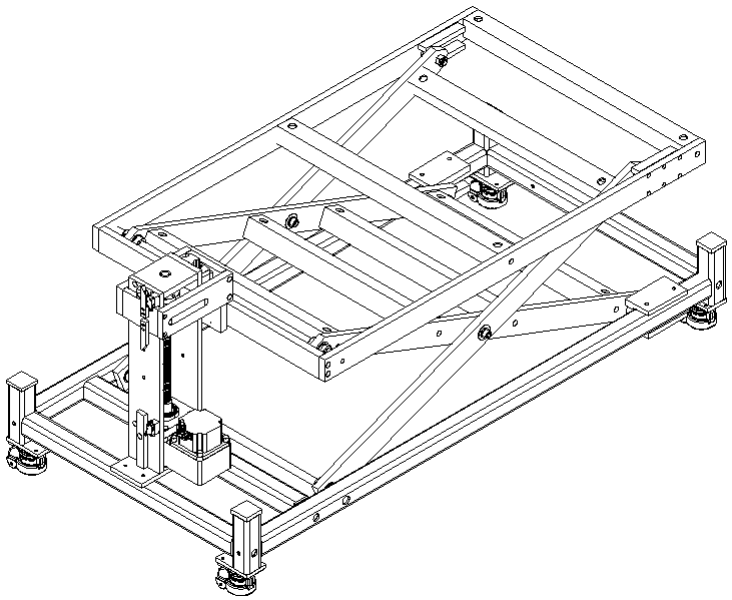
- Max Lift Hight is 22.00” inches
- Stork 9.75”
- Top platform length 30” x 49” width
- Maximum Capacity to lift concentrated central loads W1 = 15,000 N,

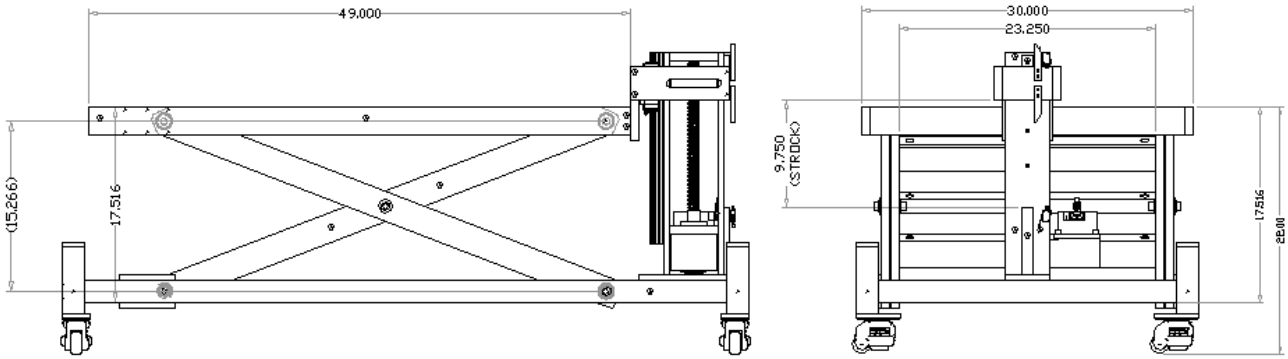
Instructions:

- This assignment Worth 10% + %1 bones mark of Overall Grade
- Show all work for every requirement and write the final conclusions in the designated zones with correct units and appropriate rounding.
- Answers outside the specified zones will not be considered.
- Marks will be deducted for incorrect notations and/or improper form.
- Submit your complete project:
  - by **midnight October 11,2026.**
  - Only in Blackboard, otherwise it will not be accepted.
  - as a single file,.
  - In PDF format.
- Submissions in formats other than those listed above will not be accepted

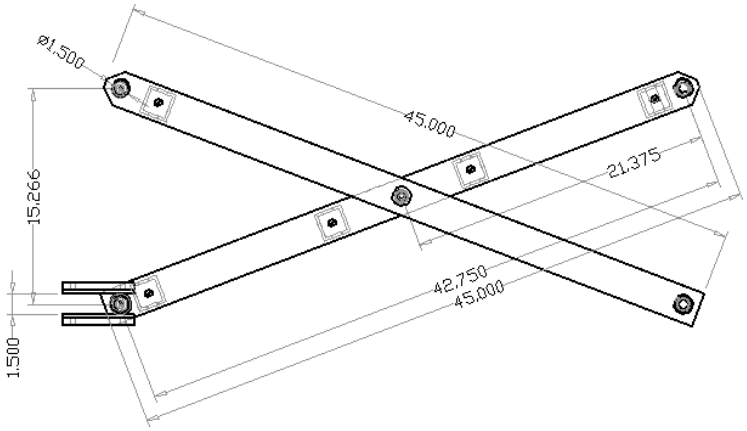
Late submission penalties:

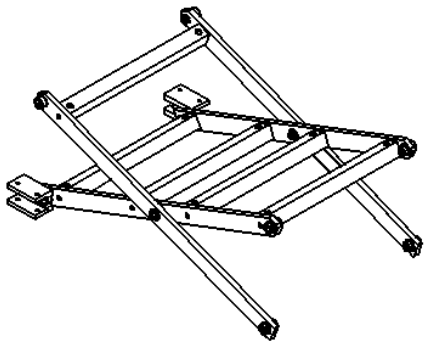
One day (-1.0 mark)	Two days (-2.0 marks)	Three days (-3.0 marks)	Four days (-4.0 marks)	Five days (-5.0 marks)





2D schematic drawing of the scissor lift at its extreme position (22)", showing the main dimensions, such as the scissor table heights, base lengths, etc.





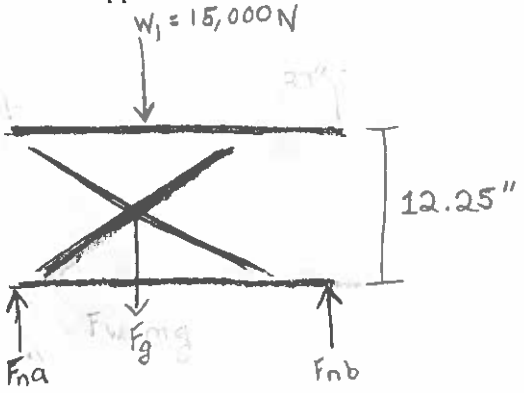
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Requirements 01: (4 marks)

A FBD sketch should include:

- **A simplified outline:** Sketch a clean, basic shape of the outer boundaries. Leave out all internal details.
- **Completely isolated:** Do not draw the walls, floors, pins, or hinges that it contacts.
- **External forces:** Include all external force vectors, starting at the exact locations where they actually push or pull on the outline, as well as moments and internal forces such as gravity.

Insert the FBD of the scissor lift at its lower height, including only the scissor bars, top platform, and lower platform. Show all applied forces and determine all reaction forces.



Lowest Height:
22" - 9.75" = 12.25"

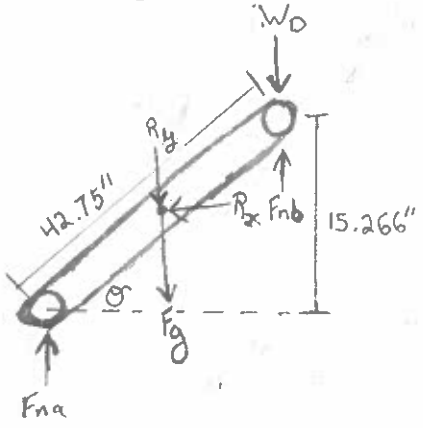
Normal Force A = 7,500

Normal Force B = 7,500

Insert the FBD of the scissor lift bar (one of the scissor sides) when the top platform is at its maximum height.

The FBD must:

- Show all applied forces and reaction forces acting on the bar.
- Determine all reaction forces.
- Clearly show the angle of the scissor bar relative to the horizontal (x-axis).



$\theta = \sin^{-1}\left(\frac{Opp}{Hyp}\right)$

$= \sin^{-1}\left(\frac{15.266}{42.75}\right)$

$= 20.9°$

$F_g = mg$

$F_{na} = F_g/2$

$F_{nb} = F_g/2$

R_y - Reaction Force y

R_x - Reaction Force x

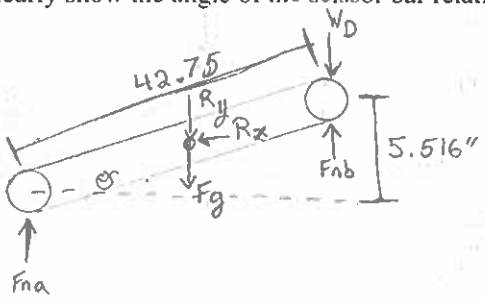
W_D - Weight Distributed

$W_D = 7,500\ N$

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Insert the FBD of the scissor lift bar when the top platform is at its lowest height.
The FBD must:

- Show all applied forces and reaction forces acting on the bar.
- Determine all reaction forces.
- Clearly show the angle of the scissor bar relative to the horizontal (x-axis).



Lowest Height:
$15.266" - 9.75" = 5.516"$

$\theta = \sin^{-1}\left(\frac{Opp}{Hyp}\right)$
$\theta = \sin^{-1}\left(\frac{5.516}{42.75}\right)$
$\theta = 7.4°$

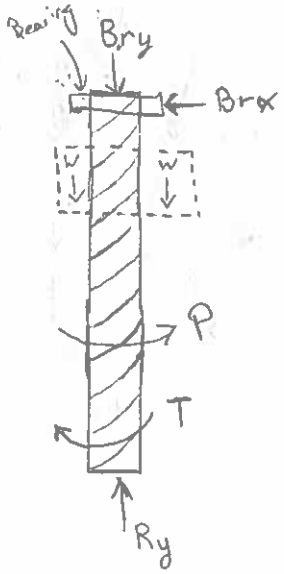
R_y - Reaction Force y
R_x - Reaction Force x

$F_g = mg$
$F_{na} = F_g/2$
$F_{nb} = F_g/2$

W_D - Weight Distributed
$W_D = 7,500N$

Insert the FBD of the Acme screw when the lift any position".
The FBD must:

- Show all forces and moments acting on the Acme screw.
- Identify the applied forces and reaction forces.
- Determine all reaction forces and moments.



T → Torque
Ry → Reaction Force y
W → Weight
Bry → Bearing reaction y
Brx → Bearing reaction x

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Requirements 02: (3 marks)

Calculate to design, verify and confirm the acme screw used for this design.

Investigate options to find;

- the friction angle ✓
- the helix angle ✓
- P; Effort applied at the circumference of the screw to lift the load of 15000N ✓
- maximum required torque, ✓
- maximum required power ✓
- calculate the efficiency. ✓
- Verify the selected gear motor in the design attached to this assignment (use Oriental motors website)

ACME:
$2\beta = 29°$
$\beta = 14.5$

Trap:
$2\beta = 30°$
$\beta = 15°$

Given:

$W = 15\,000\ N$

Stroke = 9.75" → 248 mm

Time = 90sec

Velocity = 2.756 mm/s → $\frac{2.756}{1000}$ → 0.002756 m/s

Outlet:

V = 110 VAC

I = 15 Amp

McMaster-Carr Part #:
7549K81

Lead = 5mm

Pitch = 5mm

diameter = 24mm

$h = \frac{P}{2}$

$h = \frac{5mm}{2}$

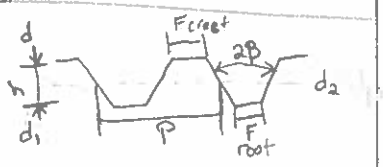
$h = 2.5mm$

Helix Angle:

$\tan\lambda = \frac{L}{\pi d_2}$

$\lambda = \tan^{-1}\left(\frac{5mm}{\pi\, 21.5mm}\right)$

$\lambda = 4.23°$



d = diameter
h = height
d_1 = Minor diameter of external thread
d_2 = Pitch diameter of external thread
P = Pitch
F = Flat width

$d_2 = d - h$

$d_2 = 24mm - 2.5mm$

$d_2 = 21.5mm$

Friction Angle:

$\phi = \tan^{-1}\left(\frac{0.15}{\cos(15)}\right)$

$\phi = 8.83°$

Using $\mu = 0.15$
Hardened steel on Cast Iron

Calculating P:

$\tan\lambda = \frac{p}{\pi d_a}$

$\tan\lambda = \frac{5mm}{\pi \cdot 21.6mm}$

$\tan\lambda = 0.074$

$\tan\phi = \left(\frac{\mu}{\cos\beta}\right)$

$\tan\phi = \left(\frac{0.15}{\cos(15)}\right)$

$\tan\phi = 0.1553$

$P = W\tan(\lambda+\phi) = W\left[\frac{\tan\lambda + \tan\phi}{1-\tan\lambda\tan\phi}\right]$

$P = 15\times10^3\left[\frac{0.074+0.1553}{1-0.074\cdot0.1553}\right]$

$P = 3479.5\ N$

Max Torque:

$T = P\cdot\frac{d}{2}$

$T = 3479.5N\cdot\frac{24mm}{2}$

$T = 41754\ N\text{-}mm$

$\downarrow \times 0.001$

$T = 41.754\ N\text{-}m$

Efficiency:

$P_o = W\tan\lambda$

$P_o = 15\times10^3\cdot0.074$

$P_o = 1110N$

$\eta = \frac{P_o}{P}$

$\eta = \frac{1110N}{3479.5N}$

$\eta = 0.32\cdot100$

$\eta = 32\%$

Max Power:

$N = \frac{Velocity}{P\cdot tan}$

$N = \frac{0.002756\ m/s}{0.005m}$

$N = 0.6512\ rev/s\cdot60$

$N = 33.1\ rpm$

$w = \frac{2\pi N}{60}$

$w = \frac{2\pi(33.1)}{60}$

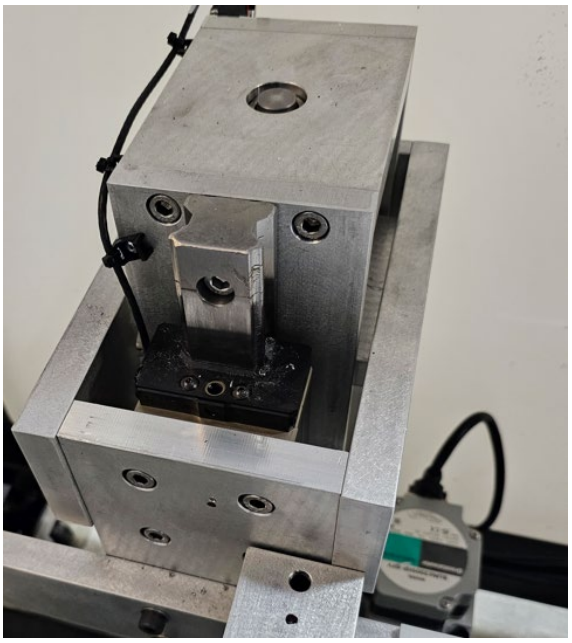
$w = 3.5\ rads/s$

Power $= T\cdot w$

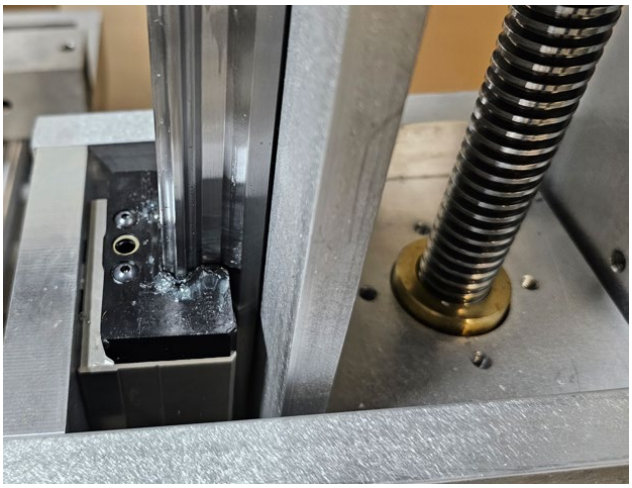
Power $= 41.754Nm\cdot 3.5\ rad/s$

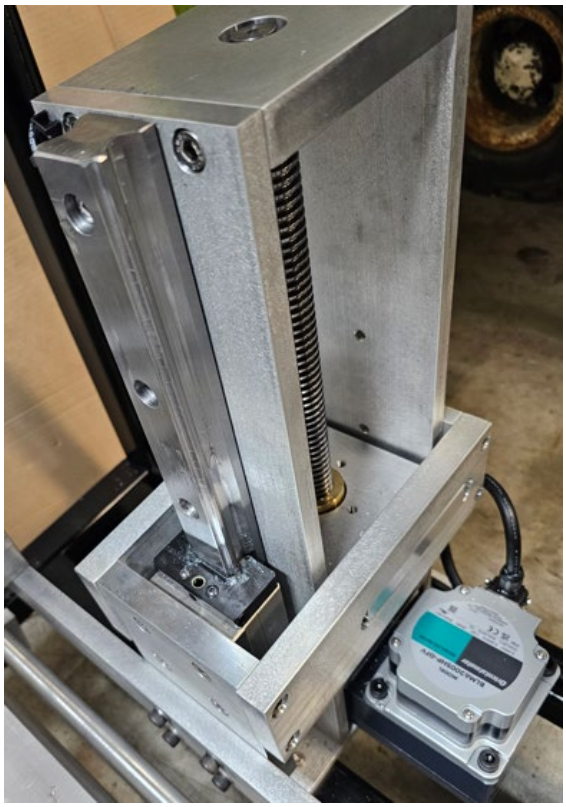
Power $= 146.14\ W$

Selected Gear Motor is Good







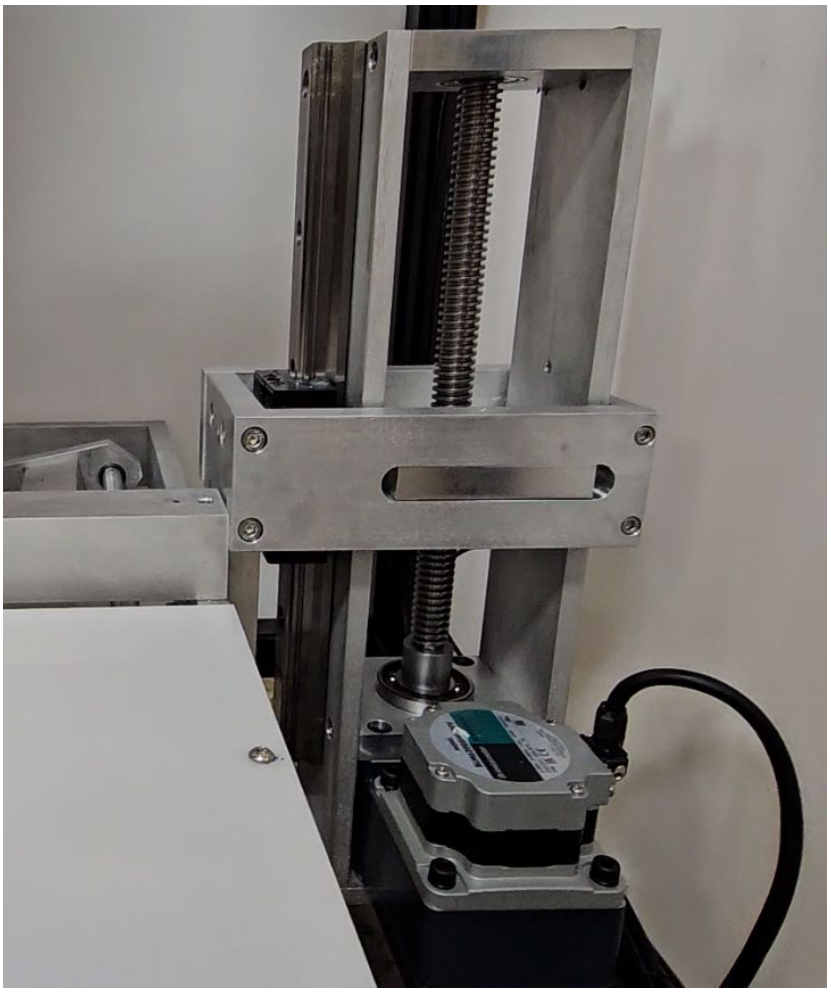












Requirements 03: (4 marks)

Based on the attached PDF containing the 2D detail drawings and design of the main actuator assembly and fabricated parts, create fully detailed manufacturing drawings for the missing components in the package, including:

- Drawing #50 — TOP BEARING PLATE
- Drawing #51 — LOWER BEARING PLATE
- Drawing #54 — NUT PLATE
- Drawing #57 — ACME SCREW

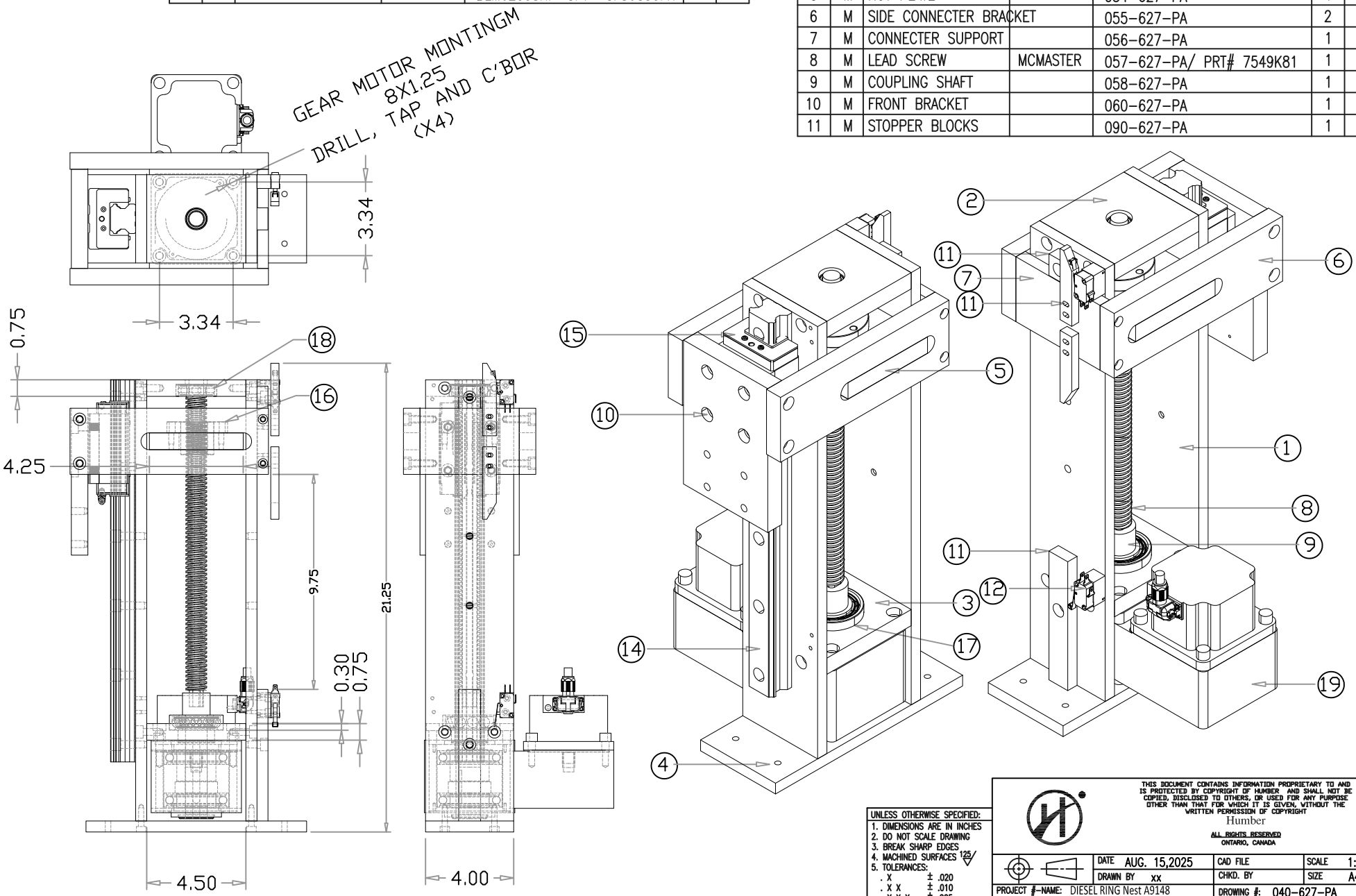
When creating the drawings, Consider the following:
Each manufacturing drawing shall contain sufficient information to manufacture and inspect the part, including:

- Appropriate projection, sufficient views, use 3rd angle projection,
- Use Humber title block, General tolerances, limits and fits and Surface finish requirements
- All required dimensions, dual dimensions where needed.
- Material specification, Heat treatment, Quantity and Manufacturing notes if required.
- Show appropriate fits and tolerances on the drawings wherever required.

12	A	MINIATURE SWITCH	McMASTER	7779K52	2	2
14	A	LINEAR BEARING RAIL	McMASTER	6709K73, 34X440mmLg	1	1
15	A	Ball Bearing Carriage	McMASTER	6709K21	1	1
16	A	FLANGE NUT	McMASTER	7549K23	1	1
17	A	ANGULAR BEARING	McMASTER	6680K27	1	1
18	A	BALL BEARING	McMASTER	5972K349	1	1
19	A	GEAR MOTOR	ORIENTAL	BLM6200SHP-GFV GFS6G30FR	1	1

BILL OF MATERIAL

ITEM	M P	DESCRIPTION	MATL/MAKER	DWG. NO./PART NO./DIM.	QTY	TOT
1	M	ACTUATOR FRONT WALL		049-627-pa	1	1
2	M	TOP BEARING PLATE		050-627-PA	1	1
3	M	LOWER BEARING PLATE		051-627-PA	1	1
4	M	BASE PLATE		052-627-PA	1	1
5	M	NUT PLATE		054-627-PA	1	1
6	M	SIDE CONNECTER BRACKET		055-627-PA	2	2
7	M	CONNECTER SUPPORT		056-627-PA	1	1
8	M	LEAD SCREW	MCMASTER	057-627-PA/ PRT# 7549K81	1	1
9	M	COUPLING SHAFT		058-627-PA	1	1
10	M	FRONT BRACKET		060-627-PA	1	1
11	M	STOPPER BLOCKS		090-627-PA	1	1



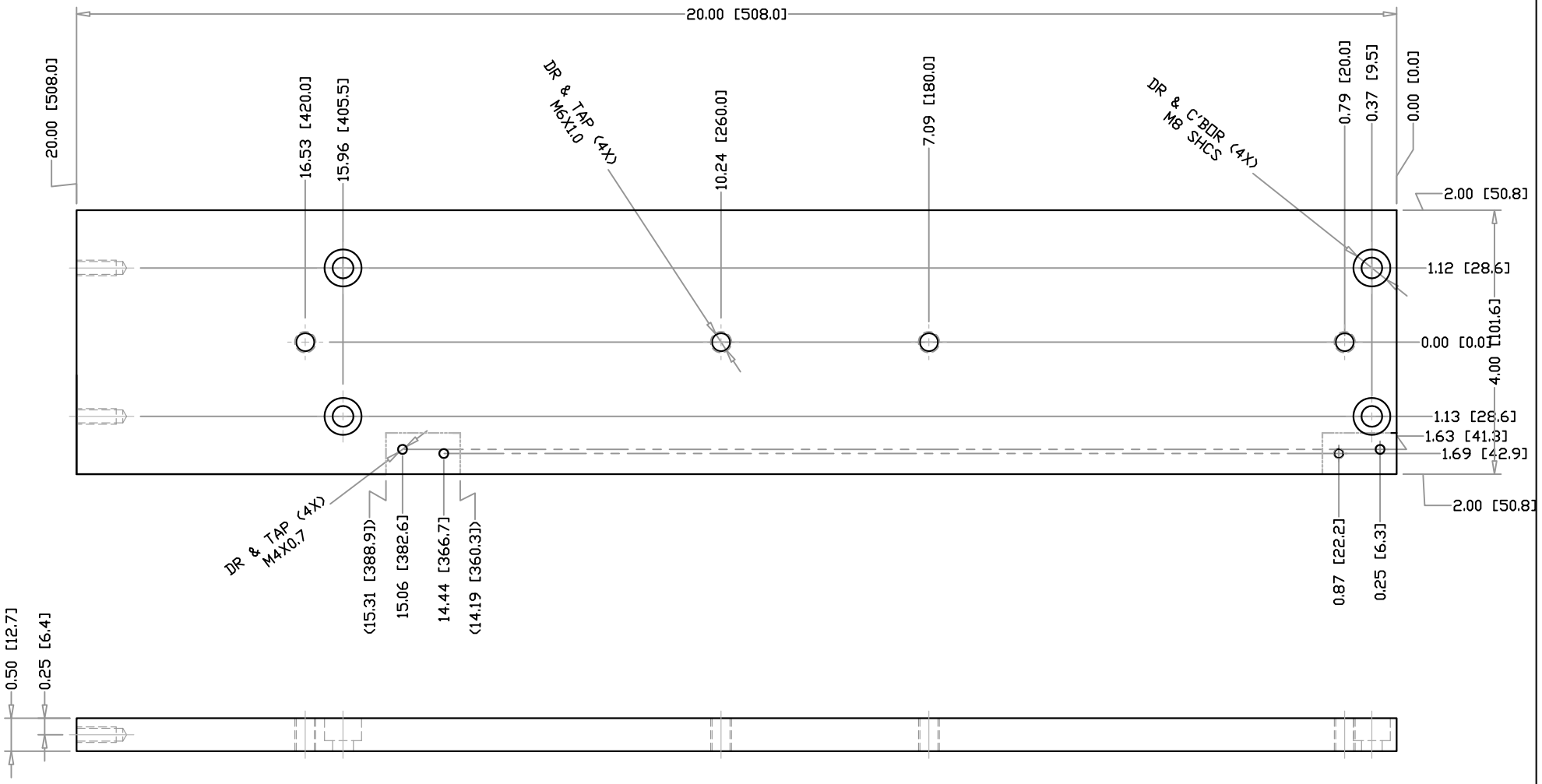
UNLESS OTHERWISE SPECIFIED:
1. DIMENSIONS ARE IN INCHES
2. DO NOT SCALE DRAWING
3. BREAK SHARP EDGES
4. MACHINED SURFACES 125
5. TOLERANCES:
. X ± .020
. X X ± .010
. X X X ± .005
FRACTIONAL ± 1/32
ANGULAR + 0' 15'

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DATE AUG. 15,2025	CAD FILE	SCALE	1:6
DRAWN BY xx	CHKD. BY	SIZE	A4
PROJECT #-NAME: DIESEL RING Nest A9148	DROWING #: 040-627-PA		
TITLE MAIN PLATE			REV. 00

BILL OF MATERIAL

ITEM	M P	DESCRIPTION	MATL/MAKER	DWG. NO./PART NO./DIM.	QTY	TOT
1	M	BAR, FLAT	6061-T	1/2 X 4.0 X 20.0	1	2

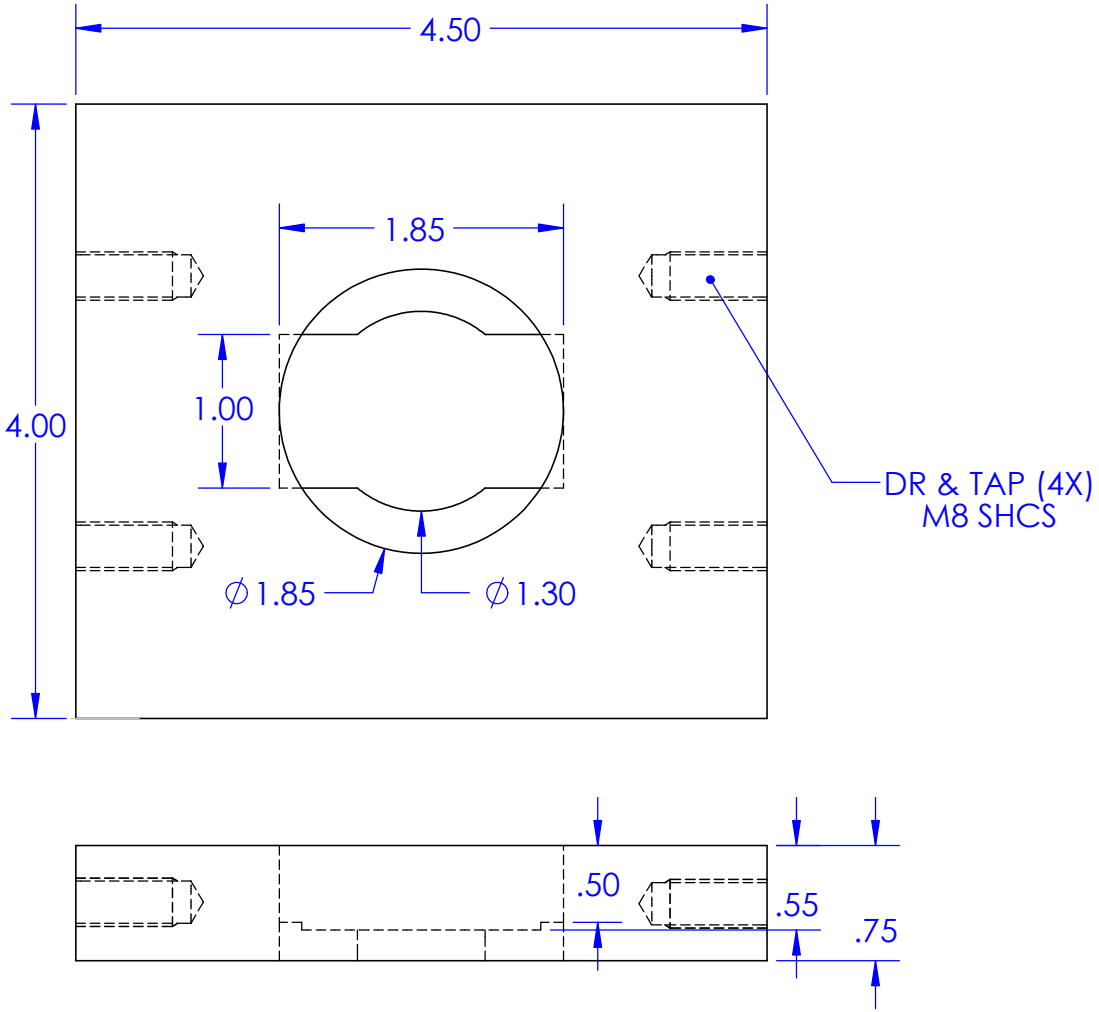


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3. BREAK SHARP EDGES
4. MACHINED SURFACES 125
5. TOLERANCES:
   . X ± .020
   . X X ± .010
   . X X X ± .005
   FRACTIONAL ± 1/32
   ANGULAR + 0° 15'

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DATE AUG. 15,2025	CAD FILE	SCALE	1:2.25
DRAWN BY xx	CHKD. BY	SIZE	A4
PROJECT #-NAME: DIESEL RING Nest A9148	DROWING #: 049-627-PA		
TITLE ACTUATOR FRONT WALL		REV.	00

ITEM NO.	PART	DESCRIPTION	QTY.
1	Top Bearing Plate	Actuator Top Bearing Plate	1



UNLESS OTHERWISE SPECIFIED:
1. DIMENSIONS ARE IN INCHES
2. DO NOT SCALE DRAWING
3. BREAK SHARP EDGES
4. MACHINED SURFACES 125
5. TOLERANCES:
. X ± .020
. X ± .020
. XX ± .010
. XXX ± .005
FRACTIONAL ± 1/32
ANGULAR ± 0 15°

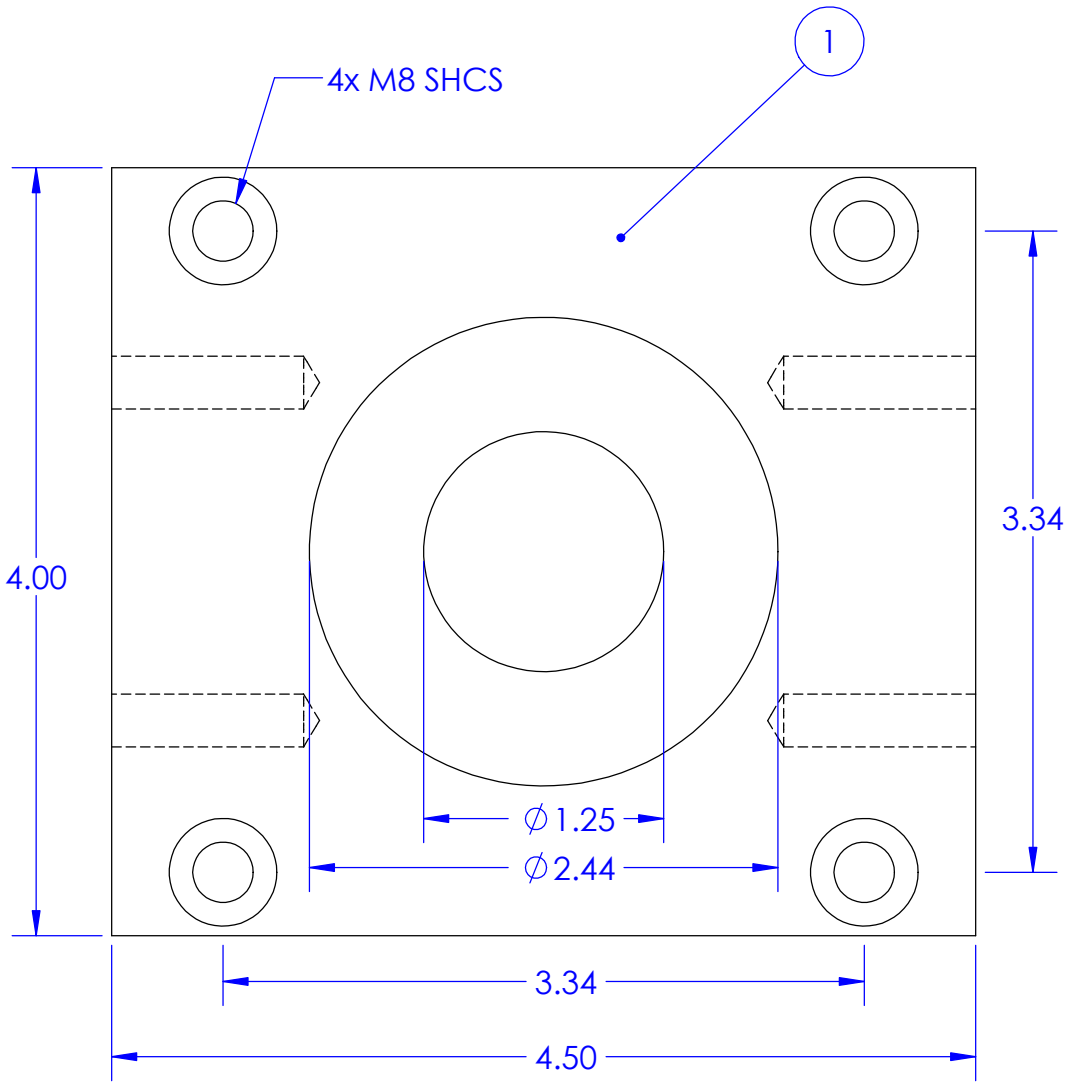
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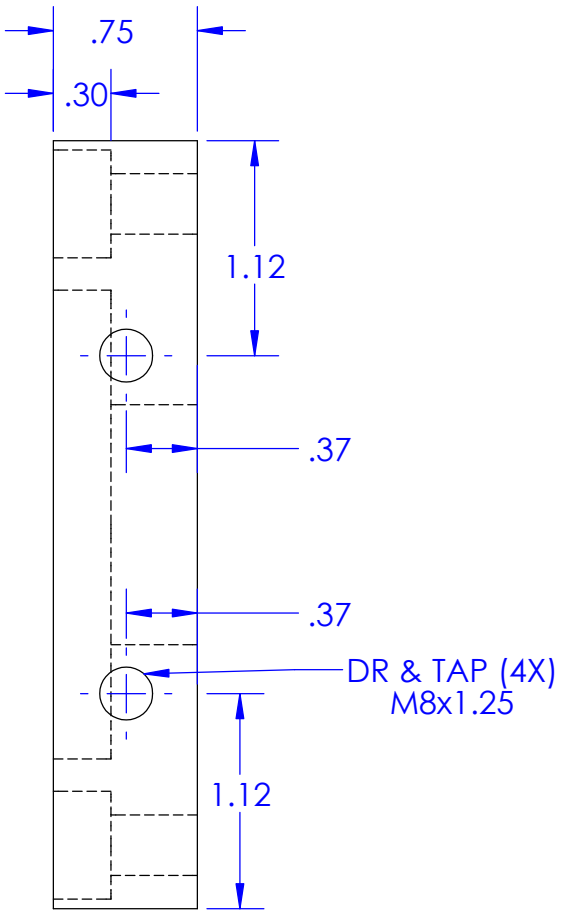
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DATE 10/05/2026
DWG # 050-627-PA
PROJECT/ ASSIGNMENT TITLE Assignment 02
DRAWING TITLE. Actuator Top Bearing Plate
REV 2
DRAWN BY Michaël McCorkell
SCALE:1:1.25

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BOM Table			
ITEM NO.	PART	DESCRIPTION	QTY.
1	Lower Bearing Plate	Actuator Lower Bearing Plate	1





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1. DIMENSIONS ARE IN INCHES
2. DO NOT SCALE DRAWING
3. BREAK SHARP EDGES
4. MACHINED SURFACES 125
5. TOLERANCES:
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. X ± .020
. XX ± .010
. XXX ± .005
FRACTIONAL ± 1/32
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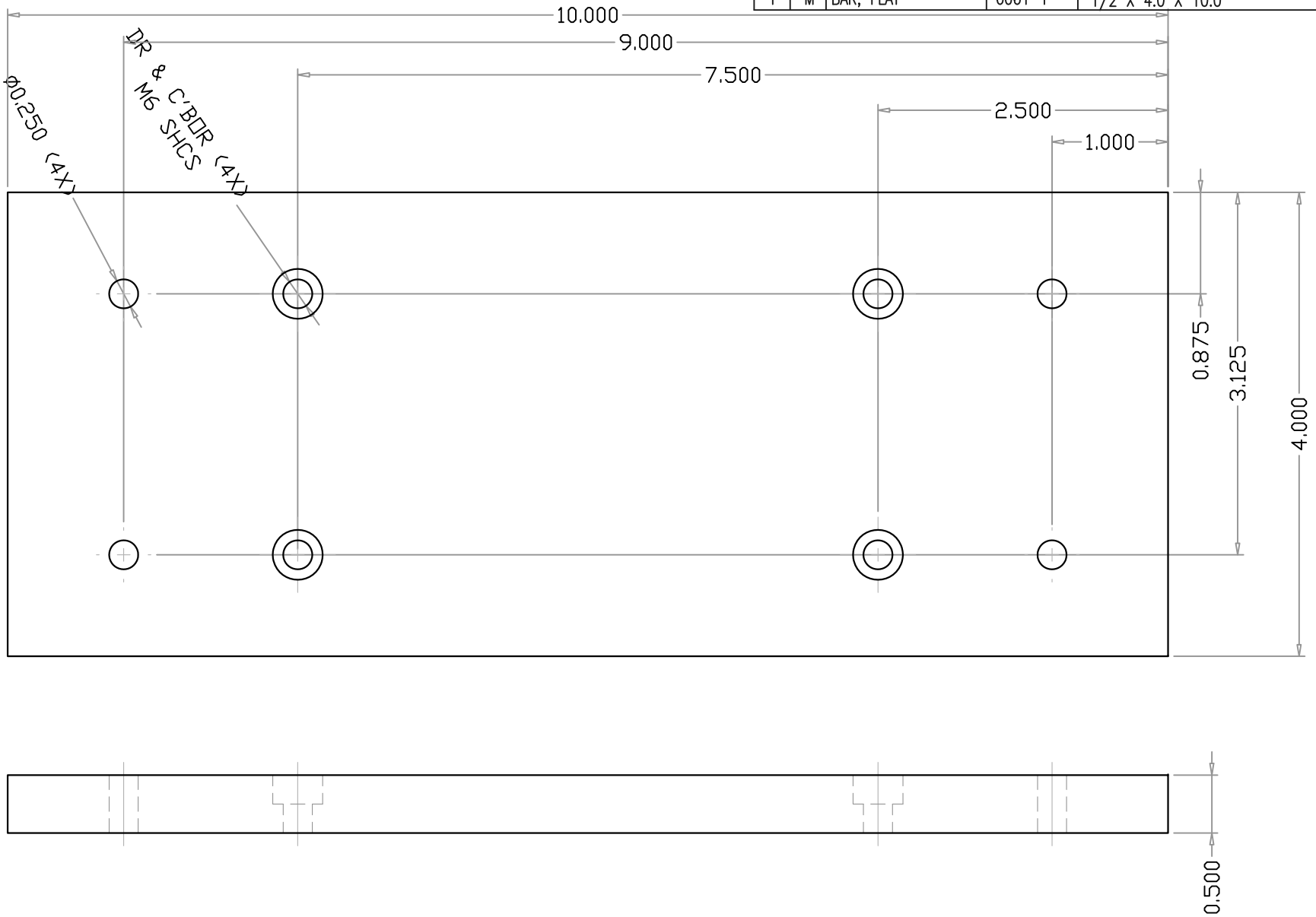
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DATE 10/05/2026
DWG # 051-627-PA
PROJECT/ ASSIGNMENT TITLE Assignment 02
DRAWING TITLE, Actuator Lower Bearing Plate
REV 1
DRAWN BY Michaël McCorkell
SCALE:1:1

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BILL OF MATERIAL

ITEM	M P	DESCRIPTION	MATL/MAKER	DWG. NO./PART NO./DIM.	QTY	TOT
1	M	BAR, FLAT	6061-T	1/2 X 4.0 X 10.0	1	1

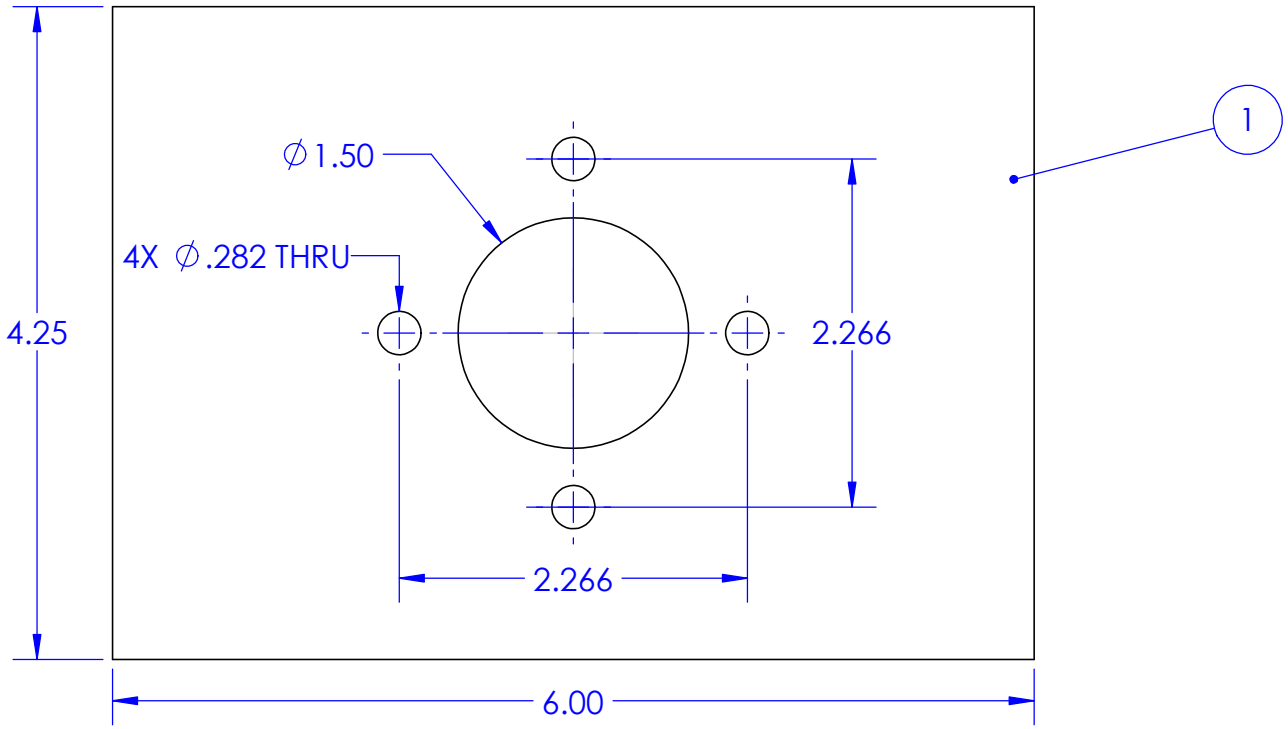


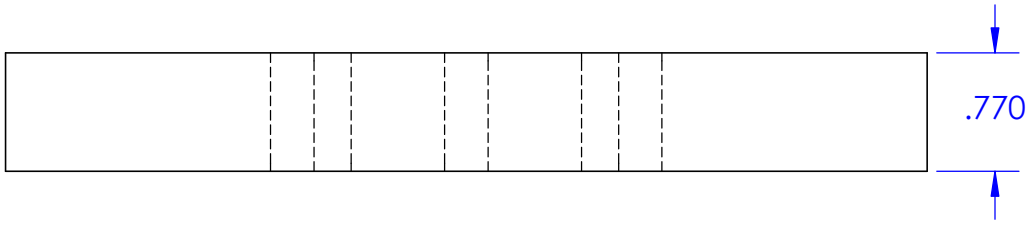
UNLESS OTHERWISE SPECIFIED:
1. DIMENSIONS ARE IN INCHES
2. DO NOT SCALE DRAWING
3. BREAK SHARP EDGES
4. MACHINED SURFACES 125
5. TOLERANCES:
   . X ± .020
   . X X ± .010
   . X X X ± .005
   FRACTIONAL ± 1/32
   ANGULAR + 0° 15'

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DATE AUG. 15,2025	CAD FILE	SCALE	1:1.25
DRAWN BY xx	CHKD. BY	SIZE	A4
PROJECT #-NAME: DIESEL RING Nest A9148	DROWING #: 052-627-PA		
TITLE ACTUATOR BASE PLATE		REV.	00

ITEM NO.	DESCRIPTION	MATL/MAKER	DWG. NO./PART NO./DIM.	QTY.
1	BAR, FLAT	6061-T	3/4 X 6.0 X 4.25	1





UNLESS OTHERWISE SPECIFIED:
1. DIMENSIONS ARE IN INCHES
2. DO NOT SCALE DRAWING
3. BREAK SHARP EDGES
4. MACHINED SURFACES 125
5. TOLERANCES:
. X ± .020
. X ± .020
. XX ± .010
. XXX ± .005
FRACTIONAL ± 1/32
ANGULAR ± 0 15°

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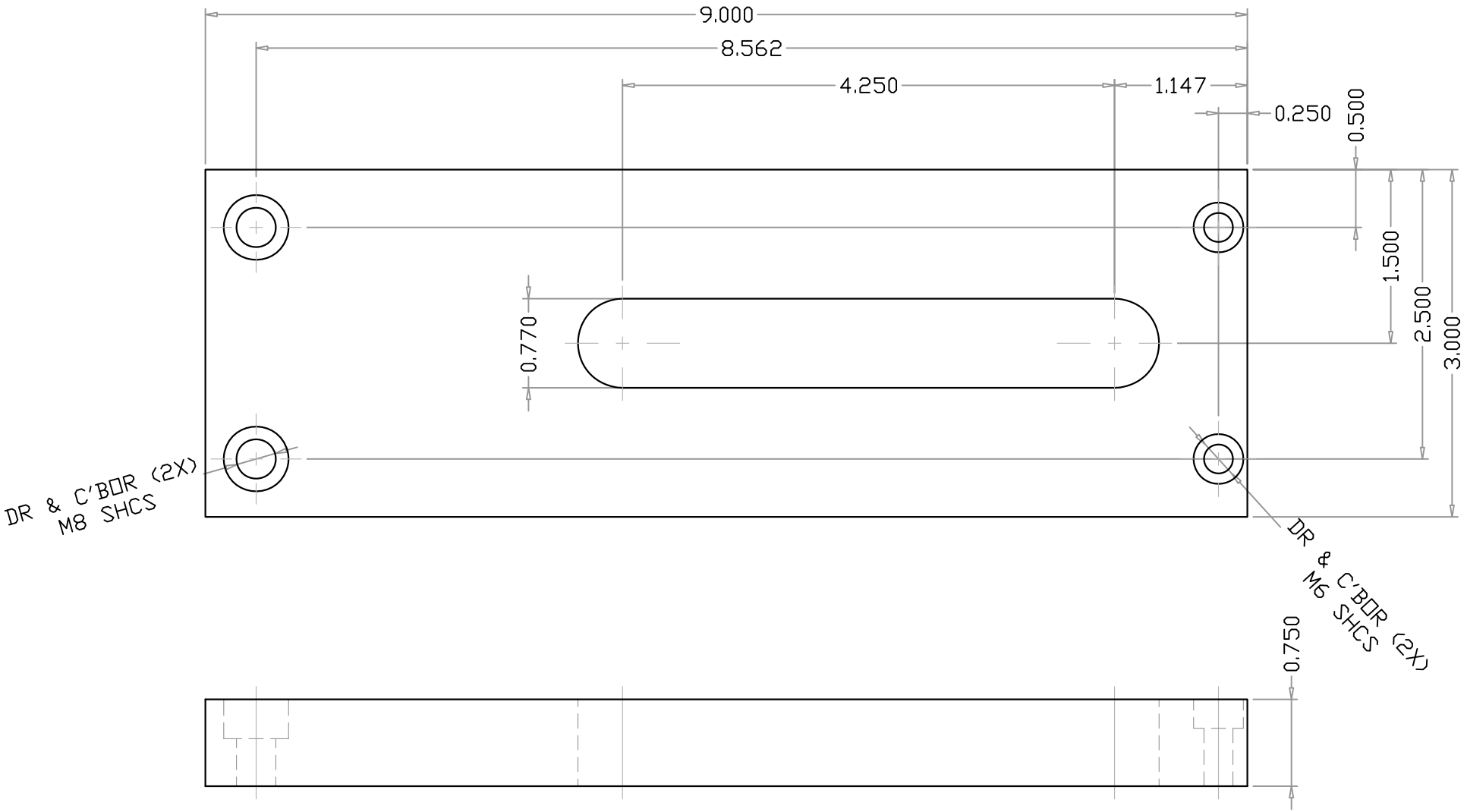
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DATE 11/08/2026
DWG # 054-627-PA
PROJECT/ ASSIGNMENT TITLE Assignment 02
DRAWING TITLE. Nut Plate
REV 2
DRAWN BY Michaël McCorkell
SCALE:1:1.25

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BILL OF MATERIAL

ITEM	M P	DESCRIPTION	MATL/MAKER	DWG. NO./PART NO./DIM.	QTY	TOT
1	M	BAR, FLAT	6061-T	3/4 X 3.0 X 5.0	1	2



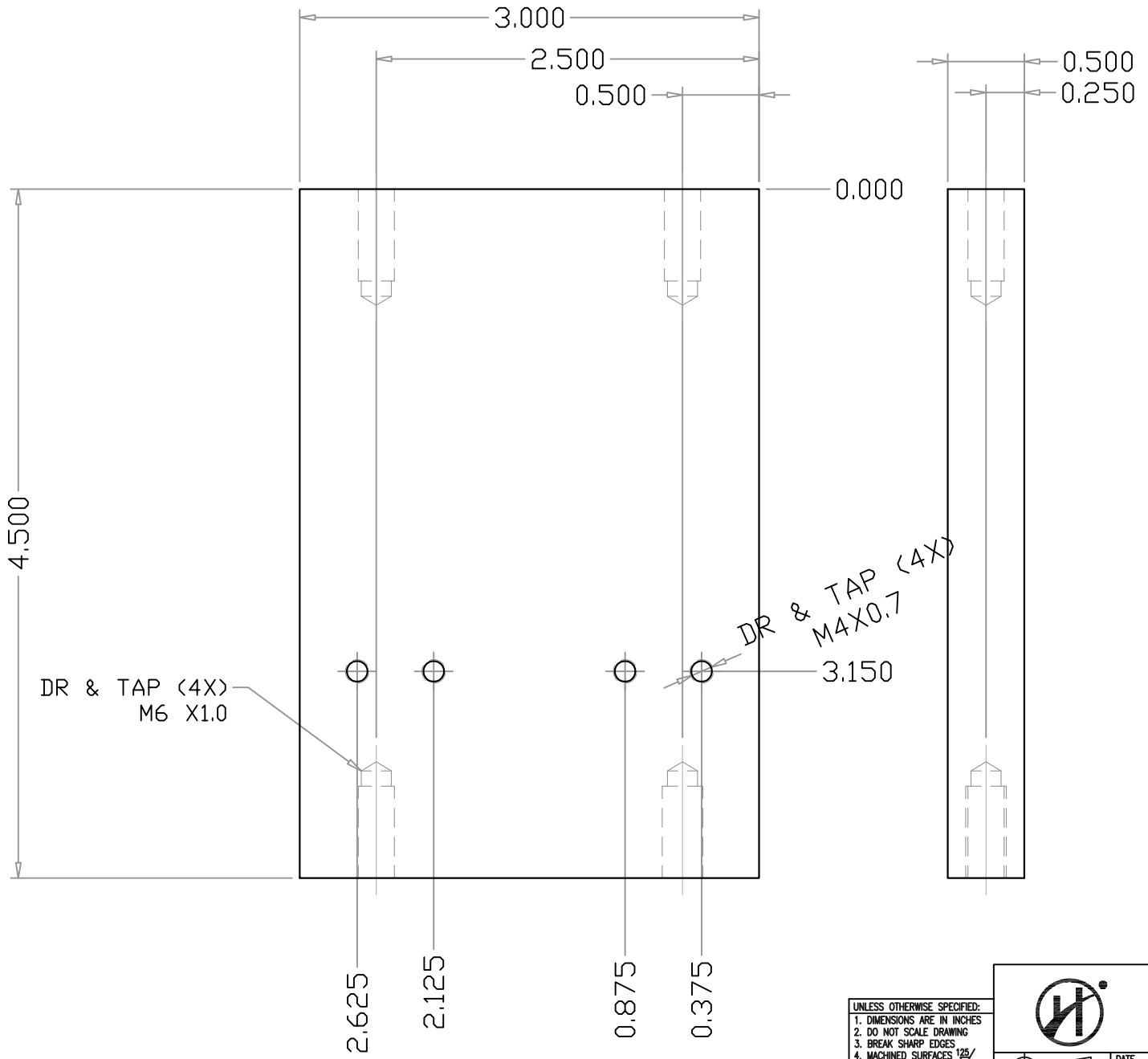
UNLESS OTHERWISE SPECIFIED:
1. DIMENSIONS ARE IN INCHES
2. DO NOT SCALE DRAWING
3. BREAK SHARP EDGES
4. MACHINED SURFACES 125
5. TOLERANCES:
   . X ± .020
   . X X ± .010
   . X X X ± .005
   FRACTIONAL ± 1/32
   ANGULAR + 0° 15'

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DATE AUG. 15,2025	CAD FILE	SCALE	1:1.25
DRAWN BY xx	CHKD. BY	SIZE	A4
PROJECT #-NAME: DIESEL RING Nest A9148	DROWING #: 055-627-PA		
TITLE ACTUATOR SIDE CONNECTER BRACKET		REV.	00

BILL OF MATERIAL

ITEM	M P	DESCRIPTION	MATL/MAKER	DWG. NO./PART NO./DIM.	QTY	TOT
1	M	BAR, FLAT	6061-T	1/2 X 3.0 X 4.5	1	1

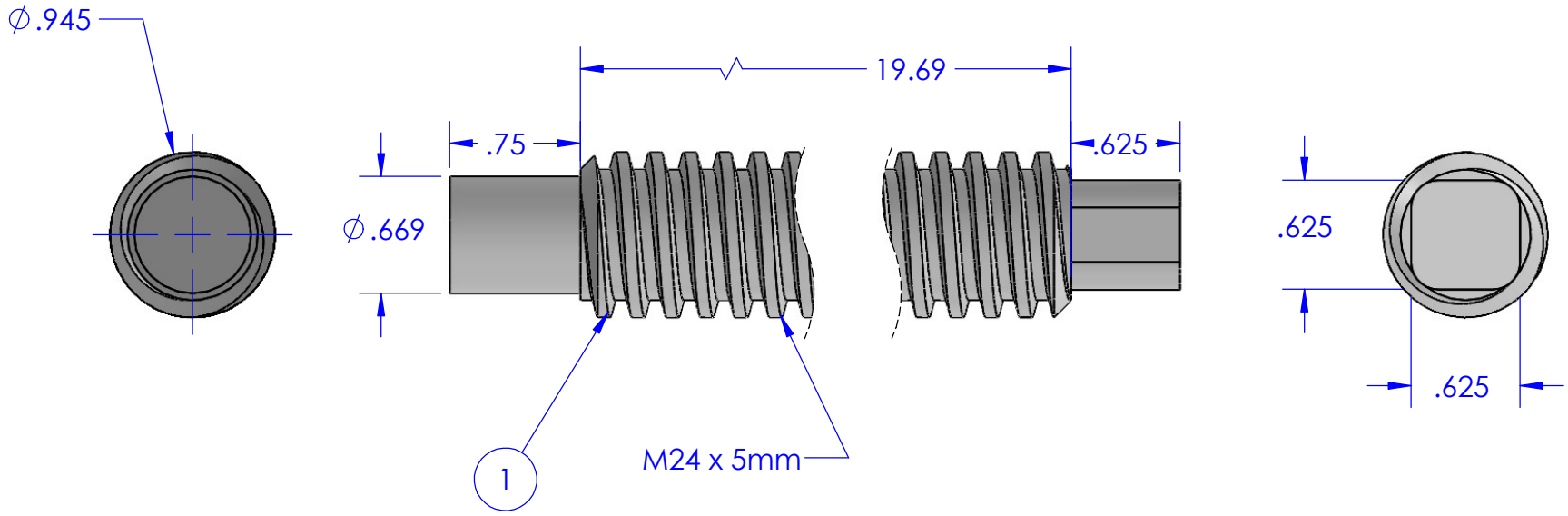


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DATE AUG. 15,2025	CAD FILE	SCALE	1:1
DRAWN BY xx	CHKD. BY	SIZE	A4
PROJECT #-NAME: DIESEL RING Nest A9148	DROWING #: 056-627-PA		
TITLE ACTUATOR CONNECTER SUPPORT		REV.	00

BOM Table				
ITEM NO.	PART NUMBER	DESCRIPTION	Mat/Maker	QTY.
1	7549K81	Lead Screw	McMaster-Carr	1



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. X ± .020
. X ± .020
. XX ± .010
. XXX ± .005
FRACTIONAL ± 1/32
ANGULAR ± 0 15°

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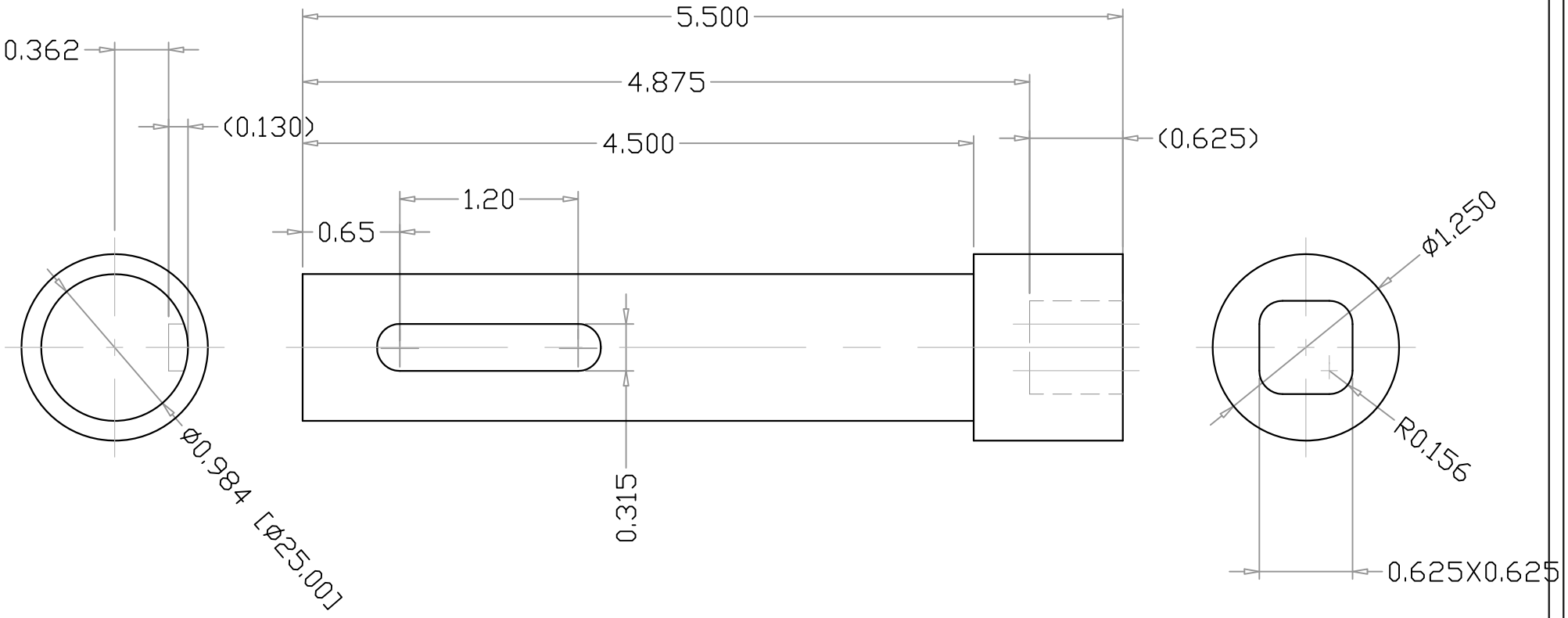
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DATE 05/10/2026
DWG # 057-627-PA
PROJECT/ ASSIGNMENT TITLE Assignment 02
DRAWING TITLE. ACME Screw
REV 1
DRAWN BY Michaël McCorkell
SCALE:1:1

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BILL OF MATERIAL

ITEM	M P	DESCRIPTION	MATL/MAKER	DWG. NO./PART NO./DIM.	QTY	TOT
1	M	BAR, ROUND	CRS	ø1.25 X 5.50	1	1



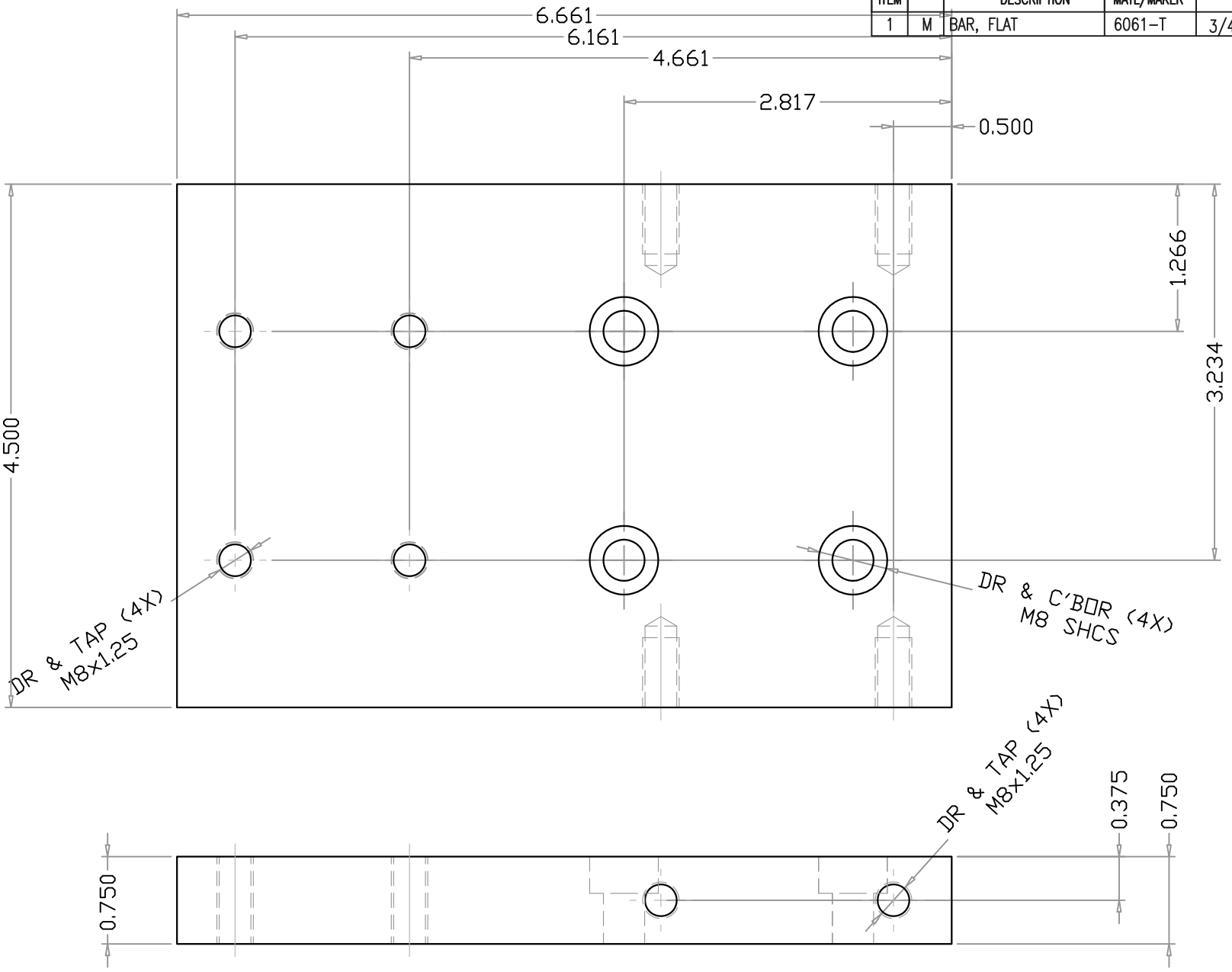
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   . X X X ± .005
   FRACTIONAL ± 1/32
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DATE AUG. 15,2025	CAD FILE	SCALE	1:1
DRAWN BY xx	CHKD. BY	SIZE	A4
PROJECT #–NAME: DIESEL RING Nest A9148	DROWING #: 058–627–PA		
TITLE ACTUATOR COUPLING SHAFT		REV.	00

BILL OF MATERIAL

ITEM	M P	DESCRIPTION	MATL/MAKER	DWG. NO./PART NO./DIM.	QTY	TOT
1	M	BAR, FLAT	6061-T	3/4 X 4.50 X 6.7	1	1



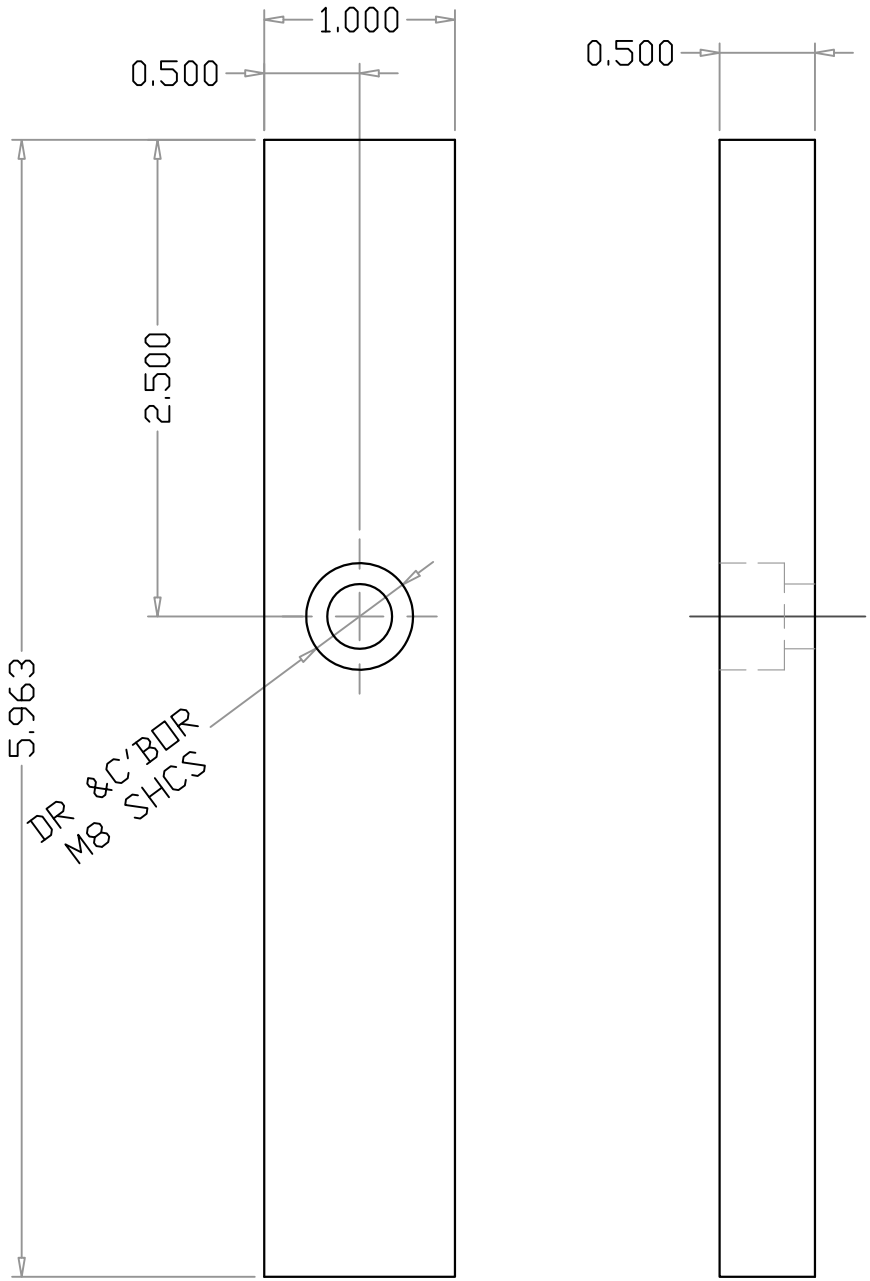
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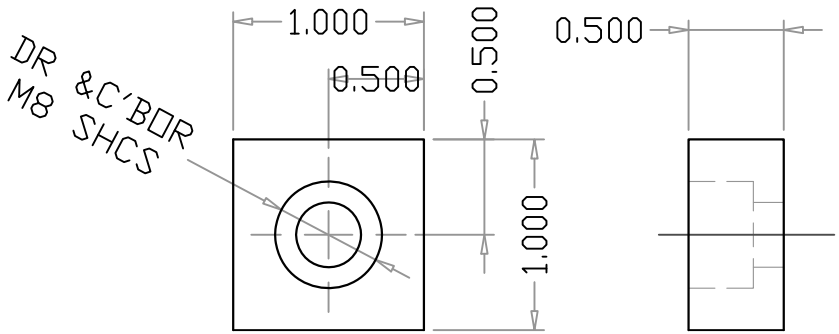
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DATE AUG. 15,2025	CAD FILE	SCALE	1:1.25
DRAWN BY xx	CHKD. BY	SIZE	A4
PROJECT #-NAME: DIESEL RING Nest A9148	DROWING #: 060-627-PA		
TITLE ACTUATOR FRONT BRACKET		REV.	00

BILL OF MATERIAL

ITEM	M P	DESCRIPTION	MATL/MAKER	DWG. NO./PART NO./DIM.	QTY	TOT
1	M	BAR,FLAT	6061-T	1/2 X 1.0 X 1.0	1	4
2	M	BAR,FLAT	6061-T	1/2 X 1.0 X 6	1	4





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DATE AUG. 15,2025	CAD FILE	SCALE	1:1
DRAWN BY xx	CHKD. BY	SIZE	A4
PROJECT #-NAME: DIESEL RING Nest A9148	DROWING #: 090-627-PA		
TITLE ACTUATOR STOPPER BLOCKS		REV.	00